

UJJAWAL PRATAP SINGH

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EDUCATION

University of Alberta, Faculty of Science
B.Sc. Computing Specialization

Edmonton, AB
Expected December 2026

Awards: University of Alberta Regional Excellence Scholarship (\$10,000)

Coursework: Web Applications & Architecture, File and Database Management, Data Structures & Algorithms, Machine Learning, Reinforcement Learning, Search & Planning in AI

EXPERIENCE

Freelance Web Developer, Kritanshi Boutique

Jul 2026 – Aug 2026

Abu Dhabi, UAE (Remote) · kritanshiboutique.com

- Shipped a production storefront as sole developer for an Abu Dhabi retail client, translating non-technical requirements into a live e-commerce site on Next.js 16 and TypeScript.
- Scoped a WhatsApp-based ordering flow that met the client's needs without payment-processor integration, removing PCI exposure from the build entirely.
- Enforced admin-only writes with Postgres Row Level Security behind a SECURITY DEFINER predicate, verified by an automated security probe (6/6 checks passing), and built an admin dashboard with product CRUD, object-storage uploads, and recoverable soft deletes that the owner runs unaided.
- Scored 99/100 Lighthouse performance on desktop with 100/100 across accessibility, SEO, and best practices, resolving WCAG contrast and ARIA violations surfaced in the audit.

Tech: Next.js 16, TypeScript, Supabase (Postgres/Auth/Storage), Row Level Security, Tailwind, Vercel

PROJECTS

Leo the Lion, AI Pediatric Health Platform

TELUS Hackathon, Top 5

- Architected a dual-interface health companion serving separate child and parent clients from a single FastAPI service, placing top 5 out of the field at the TELUS hackathon.
- Grounded model output in medical protocols with a Python RAG pipeline over an embedded vector index built on Hugging Face Sentence Transformers, serving Google Gemma 3; cut perceived latency 40% with a bi-directional Server-Sent Events layer, plus TensorFlow.js (Coco-SSD) detection and a Web Speech persona.

Tech: Python, FastAPI, Sentence Transformers, Google Gemma 3, TensorFlow.js, React 18, SSE

Social Distribution, Federated Social Network

Jan 2026 – Apr 2026

- Engineered a node-to-node federation layer delivering entries, likes, comments, and follow requests across independently deployed nodes through an inbox API secured by a custom DRF authentication backend.
- Designed a three-tier visibility model (public, unlisted, friends-only) with bidirectional follow detection enforced at both the view and serializer layers, and shipped to Heroku with PostgreSQL, Gunicorn, and Cloudinary.

Tech: Django, Django REST Framework, PostgreSQL, Heroku, Gunicorn, Cloudinary

WebAble, On-Device Accessibility Extension

Feb 2026

- Wrote a zero-dependency Multi-Layer Perceptron inference engine in vanilla JavaScript, running forward propagation and softmax classification over 63-dimensional landmark vectors with no ML runtime.
- Held 60 FPS ASL classification via MediaPipe Hands landmark extraction, and shipped live captions, text-to-speech, dyslexia-friendly reading, and epilepsy-safe modes — every inference step on-device, so no camera or gesture data leaves the browser.

Tech: JavaScript, MediaPipe Hands, Neural Network Inference, Chrome Extension APIs

Aurora, Event Lottery System (Android)

Sep 2025 – Dec 2025

- Built a native Android app resolving oversubscribed event registration with an unbiased waitlist randomization algorithm, role-based access control across Entrants, Organizers, and Admins over Firebase Firestore real-time sync, QR check-ins, and async notifications that auto-promote waitlisted users.

Tech: Java, Android SDK, Firebase Firestore, MVC

Systems Programming — RISC-V to WASM Translator, C-Ray Engine

2024 – 2025

- Built a cross-architecture binary translator mapping RISC-V branching semantics onto WebAssembly structured control flow, with LEB128 compression and a modular decoder for register-to-stack translation.
- Built a physics-based ray tracer from scratch in C with a custom linear algebra library and manual memory management, implementing BVH acceleration, Monte Carlo integration, and global illumination.

Tech: C, RISC-V, WebAssembly, Python, Linear Algebra, Makefiles

LEADERSHIP

Indian Students Association (INDSA), VP Media

Jan 2025 – Sep 2025

- Integrated generative AI workflows (LLMs, image generation) into the content pipeline, cutting drafting and design time while holding brand consistency; analyzed engagement metrics to optimize scheduling, contributing directly to a sold-out major cultural event.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, C / C99, Bash, RISC-V

AI & Machine Learning: RAG, LLM Orchestration, NLP, Computer Vision, PyTorch, TensorFlow / TensorFlow.js, Sentence Transformers, Gemma, MediaPipe, Reinforcement Learning, NumPy, Pandas

Web & Backend: React, Next.js, FastAPI, Django REST Framework, Node.js, REST APIs, Server-Sent Events, Server Actions, Tailwind, Chrome Extension APIs

Data, Cloud & DevOps: PostgreSQL, Supabase, MongoDB, Firestore, Docker, Vercel, Heroku, GCP, Git, GitHub Actions, Linux, RLS/RBAC